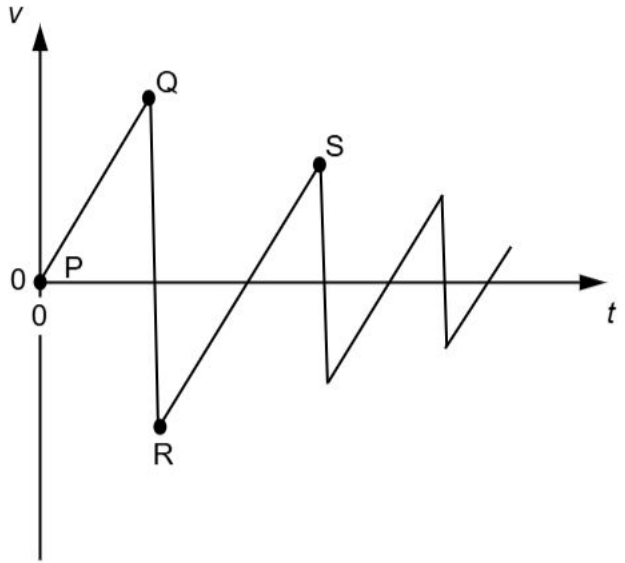
5 A ball is held above the ground and released. It falls to the ground and bounces several times.

The graph shows the variation with time t of the velocity v of the ball.



Four points on the graph are labelled P, Q, R and S.

Which statement is **not** correct?

A The area under line PQ represents the initial height of the ball.

B The collisions of the ball with the ground are inelastic.

C The gradient of the line RS represents the acceleration due to free fall.

D The maximum upwards velocity of the ball is reached at point Q.

Ans: **D**

Option A: since point P represents the point of impact with the ground, the displacement from P to Q represents the initial height it was dropped from.

Option B: Seeing that the maximum velocity between subsequent bounces gets smaller and smaller, this means that there is KE loss in the collisions.

Option C: all the positively sloped gradients represent acceleration due to free fall.

Option D: line PQ represents the ball falling to the ground, implying that the sign convention for the positive direction is downwards. Therefore, Q is the maximum downward velocity.

6 A helicopter hovers stationary in mid-air. Its rotor blades sweep out a circular path of diameter